TreeMAPPO VS Existing Methods

Q: What is 3x2+1?

Color-Advantage Mapping

-1.0 +1.0

GRPO (No Branching)

```
Q┬3x2=6. 6+1=\boxed{7} ✅
 ├3x2=42. 42+1=\boxed{43} ⚠ should not blame the last step
 ├3x2=6, 6+1=\boxed{42} ⚠ should not blame the first step
 └3x2=42, wait, actually, 3x2=6, 6+1=\boxed{7} ⚠ should not credit the first step
```

Spontaneous Branching

```
Q┬OK. Let┬'s do it step by step… ⚠ Trajectory diverges too soon
 │       └ us do it step by step…   and produce no meaningful signals
 └Let's first calculate…
```

Win-Rate or Heuristic-Based Credit Assignment

```
Q─3x2=6┬6+1=\boxed{7} ✅ High contrast for correctness divergence
       ├6+1=\boxed{42} ✅ High contrast for correctness divergence
       └6+1=7.┬Therefore, the answer is \boxed{7}. ⚠ If leaf siblings agree, the contributor
              └So, 3x2+1=\boxed{7}.                  is more likely parent than leaves
```

Our TreeMAPPO Algorithm

```
Q─3x2=6┬6+1=\boxed{7} ✅ High contrast for correctness divergence
       ├6+1=\boxed{42} ✅ High contrast for correctness divergence
       └6+1=7.┬Therefore, the answer is \boxed{7}. ✅ If leaf siblings agree, the contributor
              └So, 3x2+1=\boxed{7}.                  is more likely parent than leaves
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